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Master thesis

Enhancing Relational Understanding in CLIP Leveraging Hard Negative Captions (HNC)

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1 Introduction

In the rapidly developing field of artificial intelligence, Vision Language Models (VLMs) have demonstrated remarkable capabilities in aligning visual and textual modalities (Lu et al., 2019; Li et al., 2019; Radford et al., 2021; Li et al., 2021; 2022; Singh et al., 2022; Alayrac et al., 2022). In particular, CLIP (Radford et al., 2021) uses large-scale contrastive pretraining over millions of image-text pairs to learn a joint embedding space, yielding strong zero-shot capabilities across a wide array of tasks such as image classification, visual question answering (VQA), and image-text retrieval. A key strength of these models lies in their ability to project both image and textual inputs into a joint embedding space, enabling semantic comparisons without the need for fine-tuning on downstream tasks.

However, despite its superior performance in object recognition and broad concepts, CLIP often struggles with relational reasoning, which needs identify fine-grained spatial configurations and semantic relationships among multiple entities in a scene.

For example, as shown in Figure 1, CLIP with ViT-B/32 as vision encoder model, reports a cosine similarity score of 0.1833 between image and positive caption “The trees are behind the fence”, while reports a higher similarity score of 0.1907 between image and negative caption “The trees are in front of the fence”, treating object identities correctly but ignoring relational cues. This highlights a fundamental weakness in CLIP’s ability to understand the relationships. This relational understanding is vital for downstream tasks such as Visual Question Answering (VQA), scene graph generation, and robotic perception, especially in zero-shot scenarios where the model needs to identify the spatial position of objects.

CLIP’s limitation in relational understanding arises in part from its use of a global contrastive training objective, which encourages high-level alignment between entire image and text embeddings. While this promotes strong object-level recognition and general semantic matching, it often overlooks the internal structure of scenes, such as spatial relationships, attribute binding, and logical dependencies. As a result, the model tends to treat images and captions as unordered sets of entities,

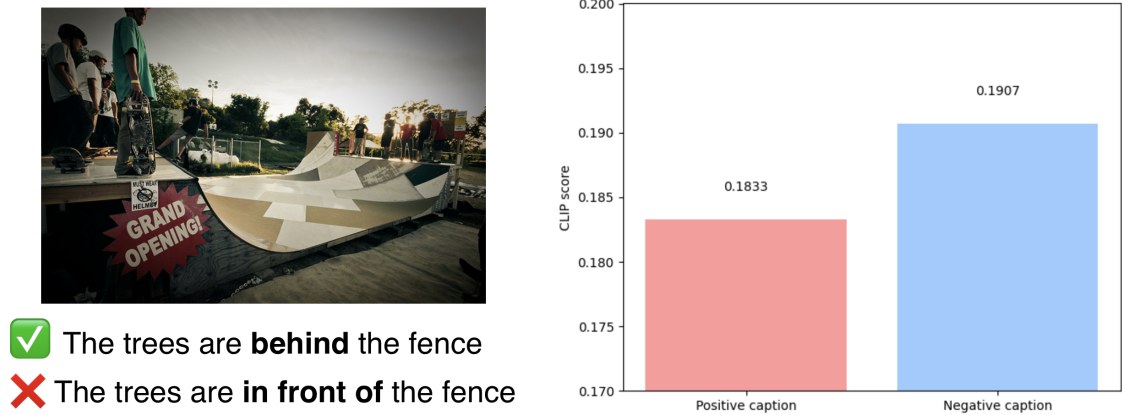


Figure 1: Example of CLIP scores between the image and positive/negative captions.

failing to capture compositional semantics and fine-grained relationships. Additional efforts are required to bridge this gap and enhance the model’s sensitivity to relational cues.

In this thesis, I propose a targeted fine-tuning strategy that leverages challenging hard negative captions (HNC) (Dönmez et al., 2023), which captions that are minimally contradictory to their paired image, generated by altering a piece of information in the corresponding positive caption. To enforce sensitivity to these subtle relational shifts, I design a contrastive loss function which can be combined with hard negative captions. For hard negative captions loss, I assign a dynamic weight, allowing explicit control over its influence during training. To preserve CLIP’s original object–text alignment capabilities and avoid catastrophic forgetting, I experiment with two separate regularization strategies: L2 regularization and a Kullback–Leibler (KL) divergence penalty, each applied independently in different variants of the model. Finally, I adapt a novel caption–image interactions pipeline (Möller et al., 2025) to visually interpret and validate the relational reasoning improvements in fine-tuned model

This thesis is organized as follows. Chapter 2 reviews relevant background of vision–language models, the contrastive learning framework of CLIP, and existing efforts to enhance its relational understanding capabilities. It also introduces the

Hard Negative Caption dataset as a semantically structured resource for fine-grained supervision. Chapter 3 presents the proposed methodology, detailing the HNC construction and integration into contrastive training, the revised loss function, and two regularization strategies to mitigate catastrophic forgetting. Chapter 4 describes the experimental setup, covering dataset, pairing strategies, implementation details, and evaluation metrics. Chapter 5 reports the empirical results, analyzing model behavior across training modes, visual similarity patterns, the impact of regularization, and the effects of hard negative weighting. Chapter 6 applies an explanation framework to visualize attribution maps. Finally, Chapter 7 concludes the thesis and outlines potential directions for future research.

2 Related Work

This chapter presents the background and related work that underpin this thesis. I begin by briefly reviewing Vision–Language Models (VLMs), with a focus on their architectural principles and relevance to image–text alignment. Next, I discuss the contrastive learning framework used in CLIP, highlighting its strengths and known limitations in relational reasoning. I then introduce prior efforts that aim to enhance relational understanding within the CLIP architecture. Finally, I describe the Hard Negative Caption (HNC) dataset, which serves as the foundation for the proposed fine-tuning approach in this thesis.

2.1 Vision–Language Models (VLMs)

Vision–language models seek to bridge visual and textual modalities by learning joint representations that support tasks such as visual question answering (Agrawal et al., 2016; Goyal et al., 2017), image–text retrieval, caption generation, and visual grounding. Early transformer-based architectures introduce varying degrees of local and global fusion. Two-stream models like ViLBERT (Lu et al., 2019) and LXMERT (Tan and Bansal, 2019) maintain separate vision and language encoders throughout, integrating region-level visual features with token-level textual features via co-attention or cross-attention layers. Single-stream transformers such as UNITER (Li et al., 2019) process both image patches and text tokens jointly in every layer, allowing deeper, fully integrated cross-modal interactions. These fusion-oriented pre-training objectives are often complemented by masked language modeling, masked region modeling, and image–text matching tasks to encourage sensitivity to local and relational cues.

By contrast, CLIP (Radford et al., 2021) adopts a purely global alignment strategy. It uses separate image and text encoders but relies exclusively on a contrastive objective over whole-scene embeddings. This design yields impressive zero-shot capabilities and efficient nearest-neighbor retrieval (Radford et al., 2021), but tends to overlook fine-grained spatial and relational information.

2.2 Contrastive Learning in CLIP

CLIP employs a dual-encoder architecture, with one encoder processing images (e.g. ViT (Dosovitskiy et al., 2021) or ResNet(He et al., 2016)) and another processing text (a Transformer (Vaswani et al., 2023; Radford et al., 2019)), projecting both modalities into a shared embedding space. During training, CLIP computes the pairwise cosine similarity between every image and every caption in each batch, normalizes these scores into probability distributions using softmax, and applies a symmetric cross-entropy loss in both image-to-text and text-to-image directions, which is an instance of the InfoNCE objective (van den Oord et al., 2019). Reward the correct image-caption pairing and penalize all other pairings in the batch (i.e., captions paired with different images act as random negative captions). A learnable temperature parameter adjusts the peakedness of the softmax distributions, controlling how strongly the model distinguishes correct pairs from incorrect ones. This loss formulation leverages all non-matching pairs within each batch, enabling CLIP to learn robust cross-modal alignments without any task-specific supervision. By training on hundreds of millions of web-sourced image-caption pairs, CLIP achieves remarkable zero-shot transfer across classification, retrieval, and other downstream tasks. However, by relying exclusively on global embedding alignment, as mentioned before, CLIP overlooks the structured dependencies between individual regions and text tokens, making it insensitive to precise spatial arrangements and semantic relationships within a scene (Yuksekgonul et al., 2023).

2.3 Relational Enhancement in CLIP

Some approaches have been proposed to address these limitations in CLIP’s ability to understand object relationships. NegCLIP (Yuksekgonul et al., 2023) introduces composition-aware hard negative mining during training. It systematically perturbs relationships within captions, such as swapping subjects and objects or altering verb phrases, without breaking grammaticality. This structured perturbation encourages the model to focus on relational semantics, thereby improving its ability to distinguish correct from incorrect relational descriptions during training. NegCLIP shows

strong gains on benchmarks like Visual Genome(VG)(Krishna et al., 2016), suggesting the model becomes more attuned to the roles and relational configurations within a sentence. In addition to text-level perturbations, NegCLIP also incorporates hard negative images to further increase contrastive difficulty. Specifically, it samples visually similar but semantically different images to act as negative image candidates within each training batch.

While this dual-level hard negative strategy enhances relational sensitivity, it also increases the complexity of the training pipeline, requiring additional preprocessing steps to compute similarity between each image.

Moreover, NegCLIP’s reliance on surface-level word order can introduce unintended noise. For example, if the phrase “black and white cows” is perturbed to “white and black cows”, the resulting caption may be treated as a hard negative despite preserving the same semantic content. This over-sensitivity to benign permutations may result in low-quality negative samples, potentially limiting overall model performance.

Building on this, Structure-CLIP (Huang et al., 2023) enhances relational understanding by integrating external structured knowledge in the form of scene graphs. To avoid trivial lexical perturbations, it constructs semantic negatives by manipulating scene graph triplets. Specifically, each caption is first parsed into a set of relational triplets of the form (subject, relation, object). The model then generates relationally incorrect but grammatically valid alternatives by exchanging the subject and object positions, producing triplets like (object, relation, subject). Similarly, for attribute triplets (object₁, attribute₁) and (object₂, attribute₂), the attributes are swapped if they differ, yielding negative samples such as (object₁, attribute₂) and (object₂, attribute₁).

For example, given the original caption “An astronaut is riding a horse”, the extracted triplet (astronaut, ride, horse) enables the construction of a semantically misleading caption: “A horse is riding an astronaut”. This approach offers a structured alternative to CLIP’s global embedding alignment by injecting relational supervision based on scene graph semantics.

However, this triplet-based strategy has notable limitations. First, it struggles with sentences that involve symmetric or bidirectional relationships. For instance, swapping “A man walks with a dog” to “A dog walks with a man” results in a sentence that is both grammatically and semantically acceptable, making it unsuitable as a hard negative.

Second, Structure-CLIP represents structured knowledge using fixed triplets, which simplifies scene semantics into rigid, atomic relations. While effective for identifying basic roles and attributes, this representation cannot adequately capture more complex linguistic phenomena.

These limitations also affect the performance of the knowledge-enhanced encoder module, which relies on such triplets to build semantic representations. Because the encoder is restricted to simplified structural inputs, it may fail to model deeper dependencies or compositional structures, reducing its effectiveness in guiding relational reasoning.

Together, these issues highlight the need for more sophisticated strategies for generating hard negatives, approaches that challenge the model’s fine-grained semantic understanding without relying solely on surface-level word swaps or structurally constrained representations.

2.4 Hard Negative Captions(HNC)

To address this challenge, the HNC (Hard Negative Captions) (Dönmez et al., 2023) framework introduces a more robust method. It leverages scene graph information (Krishna et al., 2016) from grounded GQA dataset (Hudson and Manning, 2019) to automatically create hard negative captions that are minimally contradictory to the visual content of the original image.

While both HNC and Structure-CLIP (Huang et al., 2023) operate over triplet-based representations derived from scene graphs, their perturbation strategies differ significantly in depth and semantic granularity. Structure-CLIP constructs hard negatives by swapping subject–object positions or exchanging attributes within triplets,

operations that are rule-based and structurally shallow. In contrast, HNC employs a semantically-aware perturbation framework that targets the deeper meaning encoded in visual-textual alignments. Rather than relying on surface-level positional inversions, it systematically introduces minimal but meaningful contradictions across a diverse range of semantic types.

Specifically, HNC categorizes captions into twelve types across five semantic classes: attribute-based, relation-based, counting-based, existence-based, and reasoning-based. It then applies targeted perturbations by modifying a single element of a selected triplet, while holding the rest of the semantic structure intact. For example, in a relation triplet (bowl, left of, cat), the relation “left of” may be replaced with “right of” to construct a minimally contradictory caption. “The bowl is to the right of the cat” generated as a hard negative caption of the positive sample “The bowl is to the left of the cat”. Similarly, in an attribute-based triplet (apple, color, red), the attribute “red” might be replaced with “green”. In reasoning-based settings, XOR logic structures provide even more nuanced contrast pairs. Sentence like “The cat is in front of either the door or the apples” may be negatively foiled by replacing “apples” with “curtain”, generating a caption that is valid on the surface but subtly incorrect in the visual context. These examples demonstrate HNC’s ability to go beyond superficial swaps, focusing instead on semantically meaningful alterations that target fine-grained comprehension.

To ensure the high quality of these hard negatives, HNC incorporates three complementary mechanisms. First, an ambiguity filtering stage removes caption pairs that are potentially confusing. Second, a plausible negative value sampling procedure selects substitutions from within the same scene graph while enforcing semantic type constraints, ensuring that negative captions are visually grounded and contextually coherent. Finally, a noisy negative filtering step leverages bounding-box annotations to verify that the altered relation truly contradicts the image content, filtering out false negatives caused by sparse or incomplete scene graphs.

Through this compositionally-aware and structurally validated framework, HNC generates hard negatives that challenge the model’s fine-grained relational understanding more effectively than traditional triplet-swap methods. Its design supports

a richer form of visual–linguistic alignment learning, enabling models to become more sensitive to subtle semantic distinctions beyond object recognition.

3 Method

This section presents a comprehensive and detailed methodology for fine-tuning CLIP using Hard Negative Captions (HNC), with the aim of strengthening its ability to model relational semantics. The method addresses CLIP’s known weaknesses in spatial and compositional understanding by introducing hard negatives, which are caption variants that retain grammatical plausibility while introducing subtle contradictions of the original positive captions. These carefully crafted contrasts encourage the model to become more attentive to semantic integrity and visual-grounded alignment. The approach builds upon the foundational CLIP model (Radford et al., 2021), using its dual-encoder architecture as a foundation, enhancing it with a relationally-aware fine-tuning framework using semantically subtle but visually grounded hard negatives. Specifically, the HNC dataset (Dönmez et al., 2023) is integrated into CLIP training using a modified contrastive loss function that dynamically weights hard negatives and includes regularization terms to mitigate catastrophic forgetting. This combined setup ensures that the model retains CLIP’s strong global alignment capabilities while injecting local information into the model.

The methodology spans several core components: the preparation and structuring of training data, the formulation of an HNC-aware contrastive objective and the application of carefully constructed regularization mechanisms.

3.1 Data Preparation Pipeline

The HNC dataset is organized in a JSON format where each entry corresponds to a caption associated with an image and is labeled as either a positive (label = 1) or a hard negative (label = 0). Each hard negative sample includes a reference `cpt_p_id` that links it directly to its corresponding positive caption. This supports semantically grounded contrasts and ensures that every negative caption is minimally altered from a realistic, image-grounded positive.

A distinctive aspect of the dataset is that each image is annotated with multiple positive captions describing different spatial, attribute, or logical relations in the

scene. For every positive caption, a semantically similar but visually contradictory hard negative caption is constructed, as shown in Figure 2. This enables nuanced semantic supervision across various relational categories.

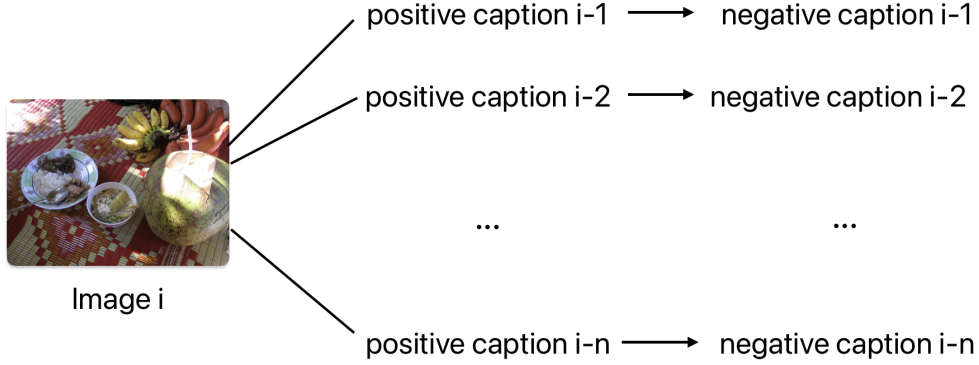


Figure 2: Illustration of image with multiple positives.

Here I pairs the data into numerous triplets (image, positive caption, hard negative caption), each capturing a distinct relational nuance of a scene. This triplet structure helps ensure that in the subsequent randomized batching process, each hard negative caption can be placed in the same batch as its corresponding positive caption. This pairing strengthens the learning signal by encouraging the model to discriminate between subtly different yet semantically connected examples.

Each triplet firstly transform into a training-ready format. For each image, the loader applies CLIP’s visual preprocessing pipeline, resizing to 224×224 , RGB conversion, and normalization with ImageNet mean and standard deviation. Simultaneously, both the positive and hard negative captions are tokenized using CLIP’s built-in tokenizer to generate text inputs compatible with the text encoder. The processed image tensor and tokenized captions are then structured into a unified dictionary containing `image_path`, `pixel_values`, `pos_text`, and `neg_text` fields.

In the context of contrastive learning, a unique challenge arises due to the multiple triplets referencing the same image. In batch-based contrastive loss computation, each image–caption pair is contrasted against all others in the batch. If a single

image appears multiple times in a batch, paired with different captions, its own positive captions may be incorrectly treated as random negatives for that image. This false penalization during training can introduce semantic noise and mislead the loss function. This is especially problematic when subtle relational distinctions are the learning target. To preserve the validity of semantic supervision, it is essential to enforce image uniqueness within each training batch.

To address this, a deduplication and refill mechanism is applied during training. As shown in Figure 3, each mini-batch is first filtered to remove entries sharing the same image path. Duplicates are identified by storing already-seen paths in a set and excluding redundant samples, ensuring that each image appears only once per batch.

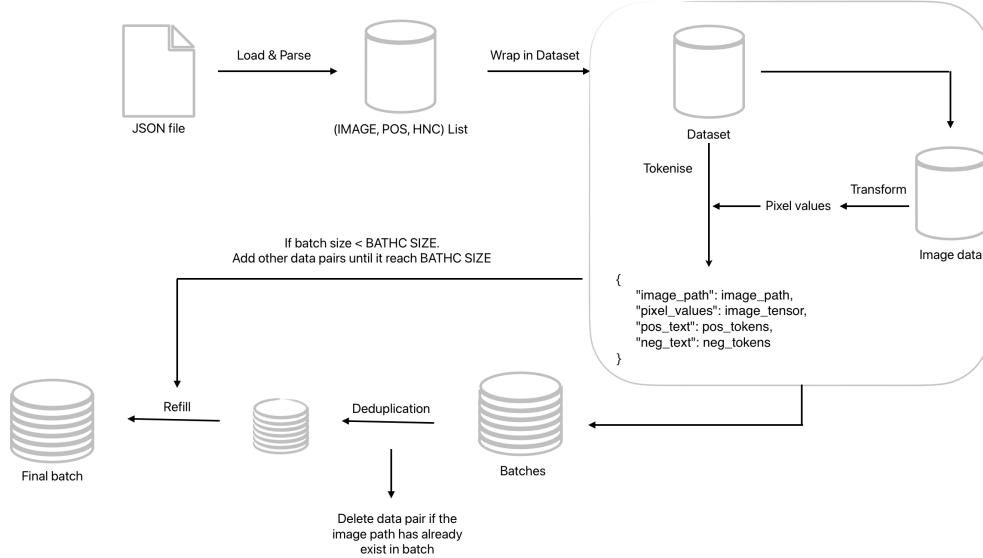


Figure 3: Data preprocessing pipeline. Deduplication and refill ensure each batch contains unique images.

If the removal of repeated images causes the batch to fall below the intended size, the refill procedure is activated. Candidates for refill are randomly sampled from the remaining dataset pool while enforcing the image uniqueness constraint. Specifically, a list of candidate indices is generated by excluding all samples whose image path already exists in the current batch. Each candidate’s image is loaded,

preprocessed, and its captions are tokenized before being appended to the batch. This process repeats until the target batch size is met.

This pipeline guarantees that all images in the final batch are visually distinct, thereby preserving the assumptions of contrastive learning and avoiding intra-image false negatives. Moreover, the new samples undergo the same preprocessing pipeline, ensuring consistency across all entries. This mechanism is especially beneficial when working with hard negatives that rely heavily on accurate image-to-caption alignment.

In summary, the data pipeline not only structures (image, positive, negative) triplets from the HNC dataset but also implements batch-level safeguards through deduplication and intelligent refill to maintain training stability and semantic clarity.

3.2 Contrastive Learning with HNC Integration

To incorporate HNC into the CLIP fine-tuning framework, a modified contrastive loss is developed to leverage the nuanced semantic contrasts provided by each (image, positive caption, hard negative caption) triplet. This section describes the construction of the HNC-aware contrastive objective and the adjustments made to the CLIP training pipeline to accommodate semantically aligned negatives.

3.2.1 Theoretical Framework

Compared to the standard CLIP setup, which primarily promotes global alignment between image and text modalities, the integration of hard negatives introduces fine-grained, local semantic distinctions. These minimally contradictory samples force the model to develop a deeper understanding of relationships, which are often overlooked during standard contrastive pretraining, while maintaining strong object recognition performance.

Figure 4 illustrates the structural extension from traditional CLIP training. The graph shows an asymmetrical similarity matrix where each row corresponds to an

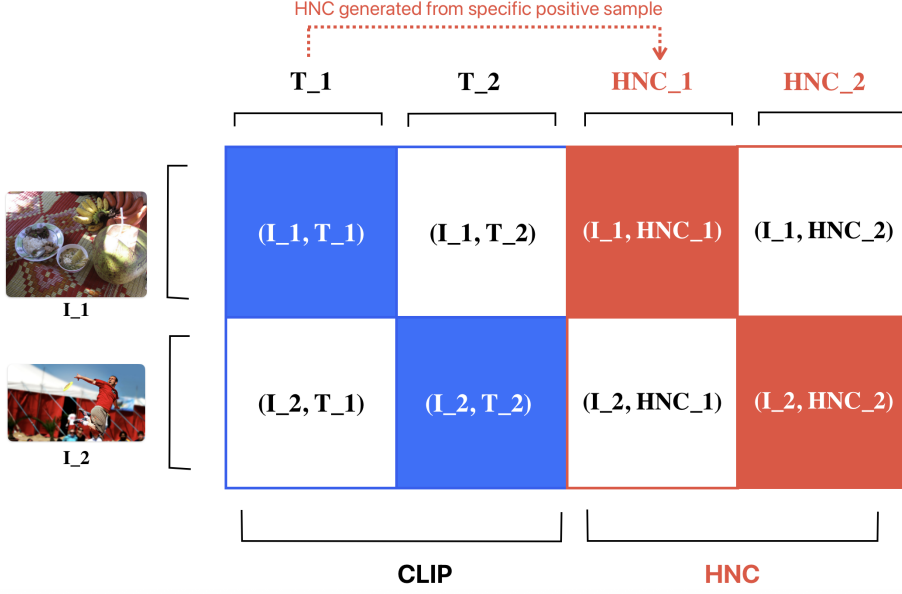


Figure 4: Combining contrastive learning with additional HNC.

image and each column to a caption. Blue cells indicate standard CLIP-style contrastive learning between an image and its matching positive caption. Red cells mark hard negatives derived from minimally perturbed versions of those positives. These hard negatives are carefully constructed to preserve grammatical fluency while introducing subtle semantic contradictions.

3.2.2 Loss Function Formulation

Formally, the image encoder f_I and text encoder f_T are used to embed image and caption pairs into a shared feature space. Given a batch of N triplets, each consisting of an image I_i , a corresponding positive caption T_i , and a hard negative caption T_i^{HNC} , the respective features are computed as $v_i = f_I(I_i)$, $t_i = f_T(T_i)$, and $\tilde{t}_i = f_T(T_i^{\text{HNC}})$. All features are ℓ_2 normalized to lie on the unit hypersphere.

Compute cosine similarities $\text{sim}(v_i, t_j)$ and $\text{sim}(v_i, \tilde{t}_j)$ between all images and both positive and hard negative captions. Here, v_i denotes the image embedding of the i -th image in the batch, while t_j represents the embedding of the j -th positive caption. The notation $\tilde{t}_j$ is used for the j -th hard negative caption (HNC). When

$i = j$, the pair (v_i, t_i) refers to a matched image-text pair, and $(v_i, \tilde{t}_i)$ refers to image with it's hard negative caption. For $i \neq j$, the terms $\text{sim}(v_i, t_j)$ and $\text{sim}(v_i, \tilde{t}_j)$ serve as random contrastive negatives from the rest of the batch. The contrastive objective includes both a standard alignment term and an auxiliary term incorporating hard negatives.

The image-to-text loss is defined as:

$$(1) \quad \mathcal{L}_i = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp\left(\frac{\text{sim}(v_i, t_i)}{\tau}\right)}{\exp\left(\frac{\text{sim}(v_i, t_i)}{\tau}\right) + \sum_{j \neq i} \exp\left(\frac{\text{sim}(v_i, t_j)}{\tau}\right) + \alpha \exp\left(\frac{\text{sim}(v_i, \tilde{t}_i)}{\tau}\right) + \sum_{j \neq i} \exp\left(\frac{\text{sim}(v_i, \tilde{t}_j)}{\tau}\right)}$$

Here, τ is a learnable temperature parameter that scales the logits, and α is a weighting coefficient that controls the influence of hard negative captions in the denominator. This formulation ensures that hard negatives contribute actively to the contrastive competition, while preserving the dominance of the matched positive pair.

The text-to-image loss retains its classical symmetric form as in CLIP:

$$(2) \quad \mathcal{L}_t = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp\left(\frac{\text{sim}(t_i, v_i)}{\tau}\right)}{\exp\left(\frac{\text{sim}(t_i, v_i)}{\tau}\right) + \sum_{j \neq i} \exp\left(\frac{\text{sim}(t_i, v_j)}{\tau}\right)}$$

The total contrastive loss is computed as the average of the image-to-text and text-to-image terms:

$$(3) \quad \mathcal{L}_{\text{total}} = \frac{1}{2}(\mathcal{L}_i + \mathcal{L}_t)$$

This loss formulation tightly integrates hard negative captions into the contrastive learning pipeline. By doing so, it enhances the model's ability to resolve subtle semantic contradictions that are grounded in visual context, improving its relational sensitivity without sacrificing its global alignment capabilities.

3.2.3 Algorithmic Implementation

Algorithm 1 Contrastive Loss with Dynamic Hard Negative Weighting

Require: I : normalized image features of batch size B

Require: $T = [T_{\text{pos}}, T_{\text{neg}}]$: positive and hard negative text features

Require: s : logit scale, t : current training step

Ensure: $\mathcal{L}$: total contrastive loss; margin value

- 1: Compute $S_{\text{pos}} \leftarrow \text{cosine_sim}(I, T_{\text{pos}}) \cdot s$
- 2: Compute $S_{\text{neg}} \leftarrow \text{cosine_sim}(I, T_{\text{neg}}) \cdot s$
- 3: **if** dynamic weighting enabled **then**
- 4: Compute w based on step t and update schedule
- 5: **else**
- 6: $w \leftarrow$ fixed value
- 7: **end if**
- 8: Initialize $W_{\text{neg}} \leftarrow \mathbf{1}_{B \times B}$
- 9: Set diagonal of W_{neg} to w
- 10: Compute weighted logits $S \leftarrow [S_{\text{pos}}, S_{\text{neg}} \odot W_{\text{neg}}]$
- 11: Define labels $y \leftarrow [0, 1, \dots, B-1]$
- 12: Compute loss $\mathcal{L}_{i2t} \leftarrow \text{CrossEntropy}(S, y)$
- 13: Compute loss $\mathcal{L}_{t2i} \leftarrow \text{CrossEntropy}(S_{\text{pos}}^\top, y)$
- 14: Compute total loss $\mathcal{L} \leftarrow (\mathcal{L}_{i2t} + \mathcal{L}_{t2i})/2$
- 15: Compute margin:

$$\text{Margin} \leftarrow \mathbb{E}[\text{diag}(S_{\text{pos}})] - \mathbb{E}[\text{diag}(S_{\text{neg}})]$$

16: **return** $\mathcal{L}$, Margin, w

To complement the mathematical formulation, Algorithm 1 presents a step-by-step pseudo-code of the proposed contrastive loss with dynamic hard negative weighting. This implementation is used during training to incorporate both the matched positive captions and their corresponding hard negatives. The hard negative weight can be dynamically adjusted over training steps to modulate difficulty, enabling the

model to first stabilize on coarse alignments before learning finer relational distinctions.

3.3 Regularization Against Catastrophic Forgetting

To ensure that fine-tuning with HNC does not disrupt the original pretrained CLIP model’s foundational capabilities, I incorporate regularization mechanisms to mitigate catastrophic forgetting (Kirkpatrick et al., 2017). Specifically, I explore two types of regularization: L2-based weight anchoring (Kirkpatrick et al., 2017) and KL divergence (Hinton et al., 2015). These are implemented independently depending on configuration, but not simultaneously.

3.3.1 L2 Parameter Regularization

The key idea of L2-based regularization is to restrict the updated model parameters from drifting too far from those of the original pretrained CLIP model, thus preserving its general representational capabilities.

This regularization requires access to the frozen pretrained weights, denoted as $\theta^{(\text{CLIP})}$, and penalizes the squared deviation of each parameter θ_i during optimization. The L2 regularization loss is formally defined as:

$$(4) \quad \text{Reg}_{L_2} = \sum_i (\theta_i - \theta_i^{(\text{CLIP})})^2$$

Then the regularization term is added to the contrastive loss to form the total objective:

$$(5) \quad \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{contrastive}} + \lambda \cdot \text{Reg}$$

Here, λ is a tunable hyperparameter controlling the strength of the regularization term.

The complete procedure for incorporating L2 regularization into training is described in Algorithm 2.

Algorithm 2 L2 Parameter Regularization

Require: θ : current model parameters

Require: $\theta^{(\text{CLIP})}$: pretrained CLIP parameters (frozen copy)

Require: λ_{reg} : regularization weight

Ensure: $\mathcal{L}_{\text{total}}$: total loss with regularization

- 1: **if** L2 regularization is enabled and $\theta^{(\text{CLIP})}$ is available **then**
- 2: Compute regularization loss:

$$\mathcal{L}_{\text{reg}} \leftarrow \sum_i (\theta_i - \theta_i^{(\text{CLIP})})^2$$

- 3: Add to contrastive loss:

$$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{contrastive}} + \lambda_{\text{reg}} \cdot \mathcal{L}_{\text{reg}}$$

- 4: **else**

- 5: $\mathcal{L}_{\text{reg}} \leftarrow 0$

- 6: $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{contrastive}}$

- 7: **end if**

- 8: **return** $\mathcal{L}_{\text{total}}$
-

3.3.2 KL Distillation from the Original CLIP Teacher

As an alternative to weight-based regularization, I adopt a knowledge distillation strategy to preserve the semantic structure of the original CLIP model. In this formulation, the original frozen CLIP model serves as a teacher, generating similarity logits between images and captions, including both positive pairs and hard negative candidates. These teacher logits are treated as a soft target distribution, and the student model is trained to mimic them via Kullback–Leibler (KL) divergence.

To soften the target distributions and highlight relative differences among candidates, I apply a temperature parameter T , following the formulation from (Hinton

et al., 2015). Specifically, for each image, both the teacher and student produce logits over $2B$ candidate captions, with B positive and B hard negative samples, which are normalized into probability distributions:

$$(6) \quad s_j = \frac{\exp(s_j/T)}{\sum_k \exp(s_k/T)}, \quad t_j = \frac{\exp(t_j/T)}{\sum_k \exp(t_k/T)}$$

The distillation objective minimizes the Kullback–Leibler divergence between the teacher distribution t and the student distribution s :

$$(7) \quad D_{\text{KL}}(t \parallel s) = \sum_{j=1}^{2B} t_j \log \frac{t_j}{s_j}$$

This KL loss encourages the student to preserve the relative ranking structure encoded by the teacher over the entire candidate set, thereby promoting semantic consistency even while learning to discriminate hard negatives.

As with the L2 regularization method, the KL loss is added to the main contrastive objective to form the total loss and λ allows flexible switching between L2 anchoring and KL-based distillation depending on the specific fine-tuning scenario.

$$(8) \quad \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{contrastive}} + \lambda \cdot \text{Reg}$$

Algorithm 3 outlines the procedure for incorporating KL divergence-based distillation.

Algorithm 3 KL Divergence Distillation from Teacher Model

Require: θ : student model parameters

Require: $\theta^{(\text{teacher})}$: frozen teacher CLIP model

Require: $x_{\text{img}}, x_{\text{text}}$: input image/text

Require: T : temperature

Require: λ_{reg} : KL regularization weight

Ensure: $\mathcal{L}_{\text{total}}$: total loss with KL distillation

```
1: Compute contrastive loss:
2:    $\mathcal{L}_{\text{contrastive}} \leftarrow \text{CrossEntropy}(S_{\text{student}}, y)$ 
3: if teacher model is available then
4:   Step 1: Get teacher embeddings
5:    $\mathbf{I}_t \leftarrow \text{Normalize}(\text{TeacherEncodeImage}(x_{\text{img}}))$ 
6:    $\mathbf{T}_t \leftarrow \text{Normalize}(\text{TeacherEncodeText}(x_{\text{text}}))$ 
7:   Split  $\mathbf{T}_t$  into positive and negative parts
8:   Step 2: Compute teacher logits
9:    $S_{\text{teacher}} \leftarrow \text{cosine\_sim}(\mathbf{I}_t, \mathbf{T}_t) \cdot s_t$ 
10:  Step 3: Compute student logits on same inputs
11:   $S_{\text{student}} \leftarrow \text{cosine\_sim}(\mathbf{I}_s, \mathbf{T}_s) \cdot s_t$ 
12:  Step 4: KL divergence loss
13:   $p_t \leftarrow \text{Softmax}(S_{\text{teacher}}/T)$ 
14:   $\log p_s \leftarrow \text{LogSoftmax}(S_{\text{student}}/T)$ 
15:   $\mathcal{L}_{\text{KL}} \leftarrow \text{KLDiv}(\log p_s, p_t) \cdot T^2$ 
16: else
17:    $\mathcal{L}_{\text{KL}} \leftarrow 0$ 
18: end if
19: Final loss:
    
$$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{contrastive}} + \lambda_{\text{reg}} \cdot \mathcal{L}_{\text{KL}}$$

20: return  $\mathcal{L}_{\text{total}}$ 
```

Both techniques aim to preserve the generalization capacity and semantic alignment encoded in the pretrained CLIP backbone, while allowing the model to adapt

to harder discriminative signals introduced by hard negative captions. The choice between L2 and KL regularization can be flexibly configured depending on the target application and experimental goals.

4 Experiments

The experiments are conducted to implement and empirically validate the proposed fine-tuning framework presented in the Method section. The primary objective is to investigate whether incorporating HNC into the CLIP training pipeline enhances the model’s sensitivity to relational semantics and its ability to capture fine-grained visual–textual alignment, while preserving the strong global matching capabilities of the original pretrained model.

This chapter introduces the complete experimental setup. It begins with a detailed description of the datasets, including the training, validation, and two contrastive test sets: one based on the human-annotated HNC benchmark, and another constructed from a MSCOCO-derived test set augmented with hard negative captions. A dedicated section outlines the pairing strategy used to convert the HNC test annotations into structured positive–negative evaluation pairs, ensuring semantic alignment and type consistency. The chapter then describes the implementation details, including model architecture, fine-tuning modes, optimizer configuration, and the integration of hard negatives through both fixed and dynamic weighting strategies. The chapter concludes with a definition of the evaluation metrics used to quantify relational sensitivity performance.

4.1 Dataset

The experiments in this work use one training set, one validation set, and two test sets, each serving a distinct role in the fine-tuning and evaluation pipeline.

The training set is constructed from the HNC dataset (Dönmez et al., 2023), by pairing each positive caption with its corresponding hard negative to form contrastive training triplets. This setup provides rich supervision by encouraging the model to distinguish between visually grounded captions that differ only in subtle relational details. For efficiency and scalability, 100k examples are randomly sampled from the full HNC training corpus for use in fine-tuning.

A separate validation set is included in the original HNC dataset, distinct from the training split. From this set, **9k** samples are randomly selected to form a held-out subset for monitoring training progress. This validation subset is used to select the best model checkpoint based on performance, instead of relying on the final training epoch.

Two test sets are used to evaluate relational sensitivity and model generalization. The first test set is provided as part of the original HNC dataset (Dönmez et al., 2023), which includes multiple caption entries per image, each labeled independently. Unlike the training data, this set is manually curated by 19 human annotators to ensure high-quality captions and minimize noise introduced by automatic generation procedures, as noted by the original authors. To support contrastive evaluation, each image in the test set is annotated with multiple individual captions, each assigned a binary label (1 for positive, 0 for negative). These are not organized as pairs by default. To convert the data into structured positive-negative pairs suitable for evaluation, a deterministic pairing strategy is applied. Details of this strategy are described in Section 4.2.

The second test set is constructed from the MSCOCO dataset (Lin et al., 2015), following the negative caption generation method proposed in Structure-CLIP (Huang et al., 2023). Hard negatives are generated by applying semantic perturbations guided by scene graphs, such as swapping relations or attributes. This COCO-derived benchmark reflects more open-domain image content, providing a complementary evaluation setting to the GQA-based HNC test set.

4.2 HNC Test Dataset Pairing Strategy

The HNC test dataset is organized at the image level, where each image is associated with multiple caption candidates. Each caption is labeled independently as either semantically correct (`label = 1`) or subtly incorrect (`label = 0`), and annotated with a semantic `type` such as `relation`, `attribute_relation`. These captions are not explicitly paired. Instead, they are presented as individual entries, each reflecting a particular linguistic perturbation or relational aspect of the image.

An example from the dataset is shown below. The image `n6309.jpg` is associated with six caption candidates. Each caption is assigned a binary label and a semantic type. In this example, captions 0 and 1 describe similar content but differ in the object referred to ("street" vs "car"); captions 2 and 3 differ in spatial relation ("behind" vs "in front of"), and so on.

```
"n6309.jpg": {
  "0": {"caption": "The people are behind the street", "type": "relation", "
    label": 0, "replaced": "street, car"},
  "1": {"caption": "The people are behind the car", "type": "relation", "
    label": 1, "replaced": null},
  "2": {"caption": "The horses are behind the trees", "type": "relation", "
    label": 0, "replaced": "behind, in front of"},
  "3": {"caption": "The horses are in front of the trees", "type": "relation
    ", "label": 1, "replaced": null},
  "4": {"caption": "The car is riding the horses.", "type": "relation", "
    label": 0, "replaced": "men, car"},
  "5": {"caption": "The men are riding the horses.", "type": "relation", "
    label": 1, "replaced": null}
}
```

While the captions appear in alternating order in this example, such structure is not guaranteed across the dataset. The captions are not explicitly grouped into positive-negative pairs, and inconsistencies such as repeated labels, missing pairs, or mismatched types can occur. This presents a challenge for evaluation. To construct meaningful contrastive pairs, a robust and consistent pairing strategy is required.

To address this, a deterministic procedure is applied. First, all caption entries for an image are sorted by their index. Then, consecutive pairs are formed in the order of appearance, such as (0,1), (2,3), (4,5). Each pair is validated to ensure it contains exactly one positive and one negative caption. Pairs with duplicate labels (e.g., both positive or both negative) are discarded to avoid ambiguity. Additionally, pairs where the two captions belong to different semantic types are excluded to preserve type consistency during evaluation.

For valid pairs, the caption with `label = 1` is treated as the positive reference, and the one with `label = 0` as the hard negative. To further ensure dataset quality, all the resulting pairs are manually checked to confirm semantic alignment and visual grounding, shown in figure 5. This filtering process reduces noise and guarantees that the evaluation set is both clean and contrastively informative. In total, the original test set contains 950 image–caption groups. After applying the pairing strategy and filtering out invalid or ambiguous pairs, 872 high-quality contrastive pairs remain for evaluation.

n143672.jpg	The white shoes are below the water bottle.	The bag is below the water bottle.	relation
n143672.jpg	People are skiing in front of the trees.	People are skiing behind the trees.	relation
n143672.jpg	People are playing on snow.	People are playing on trees.	relation
n143672.jpg	The bare trees are to the left of the crouching skier.	The red backpack is on the skier.	relation_attribute
n143672.jpg	The skier is on the white snow.	The bare trees are behind the crouching skier.	relation_attribute
n143935.jpg	The house is either behind the cows or the trees.	The house is either behind the cows or the rocks.	xor_logic_relation
n143935.jpg	There are both cows eating grass and a house behind the cows.	There are both cows eating leaves and a house behind the cows.	and_logic_relation

↓

n143672.jpg	People are skiing in front of the trees.	People are skiing behind the trees.	relation
n143672.jpg	People are playing on snow.	People are playing on trees.	relation
n143672.jpg	The bare trees are to the left of the crouching skier.	The bare trees are behind the crouching skier.	relation_attribute
n143935.jpg	The house is either behind the cows or the trees.	The house is either behind the cows or the rocks.	xor_logic_relation
n143935.jpg	There are both cows eating grass and a house behind the cows.	There are both cows eating leaves and a house behind the cows.	and_logic_relation
n143935.jpg	There is no cow that is flying on the grass.	There is at least one cow that is flying on the grass.	verify_object_relation

Figure 5: Illustration of the manually checking test pairing process.

4.3 Implementation Details

All experiments use the CLIP model with the ViT-B/32 vision encoder (Radford et al., 2021). CLIP consists of two separate components: a vision encoder for processing images and a text encoder for processing captions. Both encoders produce feature vectors that are projected into a shared embedding space, enabling direct comparison between image and text representations.

The vision encoder is a Vision Transformer (ViT) that first divides each input image into non-overlapping 32×32 patches. Each patch is converted into a vector and added with positional information. These vectors, along with a special [CLS] token, are then processed by a 12-layer Transformer. After processing, the output

corresponding to the [CLS] token serves as a summary of the entire image. This summary vector is passed through a trainable linear layer, which transforms it into a feature that lies in the same space as the text features. This allows CLIP to directly compare image and caption representations using cosine similarity.

The text encoder in CLIP is also built using a 12-layer Transformer. It first takes a natural language caption and converts each word or subword into a vector using a token embedding layer. To retain the order of the words, positional information is added to these vectors. The sequence is then passed through the Transformer, which captures the contextual meaning of each word based on its surrounding words. Among the outputs, the first token in the sequence, typically a special [SOS] token, is treated as a summary of the entire sentence. This summary vector is then passed through a learnable linear layer that maps it into the same embedding space as the image representation.

The model is initialized from the publicly released CLIP checkpoint² and fine-tuned using a distributed training setup based on DeepSpeed (Rasley et al., 2020).

To control which parts of CLIP are updated during fine-tuning, several parameter selection modes are supported. These include `text_encoder`, which updates only the text encoder and text projection layers; `vision_encoder`, which enables updates to the visual encoder and visual projection only; `full_encoder`, which fine-tunes all model components; and `last_encoder`, which restricts updates to the final transformer blocks of both encoders. Unless otherwise specified, the `full_encoder` configuration is used in all experiments. An overview of the parameter update modes and their corresponding trainable modules is provided in Table 1.

Training is performed using DeepSpeed with data parallelism. Automatic mixed precision is disabled. The effective global batch size is set to 128, using a micro-batch size of 32 per GPU and gradient accumulation over 2 steps. Gradient clipping is applied with a threshold of 1.0. ZeRO (Rajbhandari et al., 2020) Stage 2 optimization is enabled to reduce memory usage and improve scalability, while gradient checkpointing is disabled in all experiments.

²<https://github.com/OpenAI/CLIP>

Mode	Text Encoder	Text Projection	Visual Encoder	Vision Projection
<code>text_encoder</code>	✓	✓	–	–
<code>vision_encoder</code>	–	–	✓	✓
<code>full_encoder</code>	✓	✓	✓	✓
<code>last_encoder</code>	last block	–	last block	–

Table 1: Fine-tuning modes and trainable components in CLIP.

The model is fine-tuned for 20 epochs on 100k training samples. AdamW is used as the optimizer with an initial learning rate of 5×10^{-6} , weight decay of 1×10^{-4} , and betas set to (0.9, 0.98). The learning rate follows a cosine decay schedule with 100 warm-up steps.

Hard negative captions are incorporated into the training loss using a weighting strategy that controls their contribution relative to random negatives. By default, a fixed hard negative weight of 1.0 is used during training, which can be adjusted by setting the `hard_neg_weight` parameter. Alternatively, a dynamic weighting schedule can be enabled by setting `dynamic_weight` to `true`. In this case, the hard negative weight increases linearly from a minimum value (e.g., `min_weight` = 1) to a maximum value (e.g., `max_weight` = 10) over a predefined number of steps. The update schedule is controlled by `update_interval` and `num_updates`. This gradual increase allows the model to first learn basic image–text alignment and progressively emphasize subtle relational distinctions.

To prevent catastrophic forgetting, as mentioned in the previous section, two regularization methods are explored separately: L2 regularization, which penalizes deviation from the original CLIP parameters, and KL divergence, where a frozen CLIP model is used as a teacher to guide the output distribution. These methods can be activated by setting the training mode to `hnc_l2` or `hnc_kl`, respectively. The regularization strength in both cases is controlled by the hyper parameter λ , which is set to 5×10^{-6} in these experiments.

4.4 Evaluation Metrics

Evaluation is conducted in a batch-wise manner, with a batch size of 32. Each batch consists of image–positive-negative triples, where every image is associated with one positive caption and one carefully constructed hard negative caption. To assess the effectiveness of fine-tuning process, multiple types of intrinsic evaluation metrics are computed.

The first stage of evaluation focuses on positive and random negative captions. For each batch, image and positive caption embeddings are computed and normalized. The cosine similarity between each image and its corresponding positive caption is recorded, and the average similarity across all samples is used as the positive alignment score. To construct the random negative baseline, all positive captions within the batch are collected and randomly shuffled. Each image is then paired with a caption from another image, ensuring no image is matched with its original caption. The cosine similarities from these mismatched pairs are also averaged to compute the random negative score. A successful fine-tuning process is expected to yield higher positive similarity score and lower random negative similarity score, reflecting improved ability to distinguish relevant from irrelevant captions.

In addition to random mismatches, evaluation also measures sensitivity to subtle semantic errors. The cosine similarity between each image and its paired hard negative caption is computed. The difference between the positive score and hard negative score is referred to as the *margin*, which serves as an indicator of the model’s fine-grained semantic discrimination. A larger margin reflects the model’s ability to detect subtle semantic errors and distinguish correct captions from misleading alternatives.

To further quantify this contrastive separation, a threshold-based accuracy metric is introduced. For each test sample, the ratio of positive to hard negative similarity is computed. A prediction is considered correct if the ratio between the image–positive and image–negative similarities exceeds a threshold τ :

$$(9) \quad \frac{s_{\text{pos}}}{s_{\text{neg}}} > \tau$$

Accuracy is reported across multiple thresholds (e.g., $\tau = 1.0, 1.1, 1.2, 1.5, 2.0, 3.0$), with higher thresholds indicating greater confidence in semantic distinction.

All evaluations are conducted on two test datasets introduced in Section 4.1: a human-annotated HNC test set and a MSCOCO-based dataset. Both contain carefully paired contrastive caption pairs covering a range of semantic perturbations.

5 Results and Analysis

This section presents a comprehensive analysis of how various training strategies impact the model’s ability to align images with semantically correct captions and distinguish them from visually plausible but semantically incorrect alternatives. The comparison spans multiple aspects, including fine-tuning parameter modes (e.g., text-only, vision-only, full model), regularization methods (L2 vs. KL), and strategies for weighting hard negative captions during training.

Evaluation follows the metrics introduced in Section 4.4, including average cosine similarities, margin scores between positive and hard negative captions, and threshold-based accuracy across multiple semantic confidence levels. These metrics together assess both global image–text alignment and fine-grained relational discrimination.

The analysis begins by comparing training behaviors across different parameter update configurations to assess which parts of the CLIP architecture is important to relational learning. It then contrasts L2 and KL regularization techniques to evaluate their effectiveness in preserving pretrained knowledge while enhancing relational sensitivity. Next, the effect of varying the hard negative weight, including fixed and dynamic strategies, is examined to understand how loss scaling influences learning difficulty and stability.

To evaluate generalization beyond the GQA-derived HNC domain, the best-performing models are tested on a COCO-based dataset with independently generated hard negatives. Finally, a fine-grained breakdown by semantic category is performed using the annotated HNC test set, analyzing model performance across reasoning types such as spatial relations, logic operators, attribute verification, and counting.

5.1 Behavior Across Different Training Parameter Modes

To explore how different parts of the CLIP architecture contribute to learning fine-grained relational alignment, four training parameter configurations are com-

pared: training only the text encoder (**text_encoder**), only the vision encoder (**vision_encoder**), only the final layer block in both encoders (**last_encoder**), and training the entire model (**full_encoder**). Each model was trained for 20 epochs using identical optimization settings and the same HNC training data.

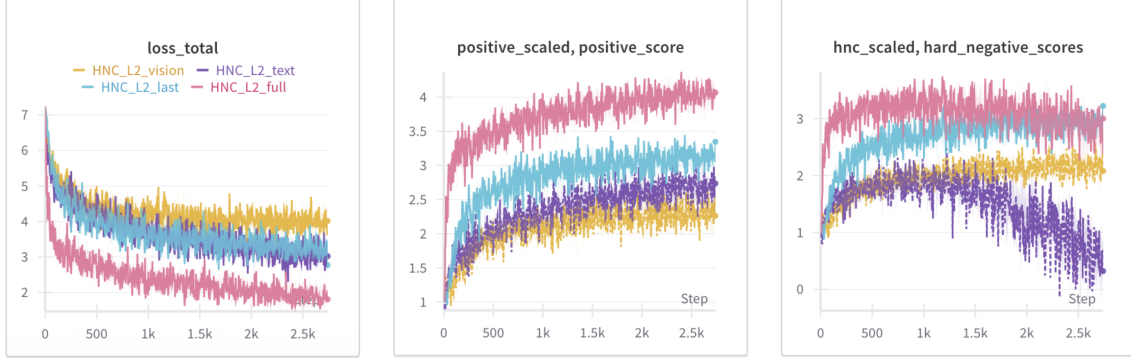


Figure 6: Training curves across different training parameter modes. From left to right: total training loss, positive similarity score, and hard negative similarity score.

Figure 6 compares the training behavior of four training parameter configurations. All models exhibit a rapid drop in total loss during the early training steps, followed by a steady plateau. This indicates that all configurations are able to fit the training data efficiently in the early stages.

The center panel shows the growth of positive similarity scores over time. All four models get improved, with the **full_encoder** (pink) configuration showing the largest increase and highest final score. This indicates that allowing updates to all CLIP layers leads to the most effective image-text alignment. The **text_encoder** (purple) and **last_encoder** (blue) variants also improve, though to a lesser extent, while the **vision_encoder** (yellow) trail behind.

The right panel tracks similarity to hard negative captions. Notably, the **vision** and **last** encoder show an initial increase in hard negative scores, which then level off. This trend suggests that these models struggle to further suppress semantically similar distractors after an early stage of learning. In contrast, the **full_encoder** and especially the **text_encoder** exhibit a distinct reversal: after an initial rise, their hard negative scores decline, indicating growing ability to suppress semantically incorrect

captions. These trends suggest that the text encoder plays a crucial role in learning fine-grained relational distinctions, while tuning only the visual encoder provides limited improvement.

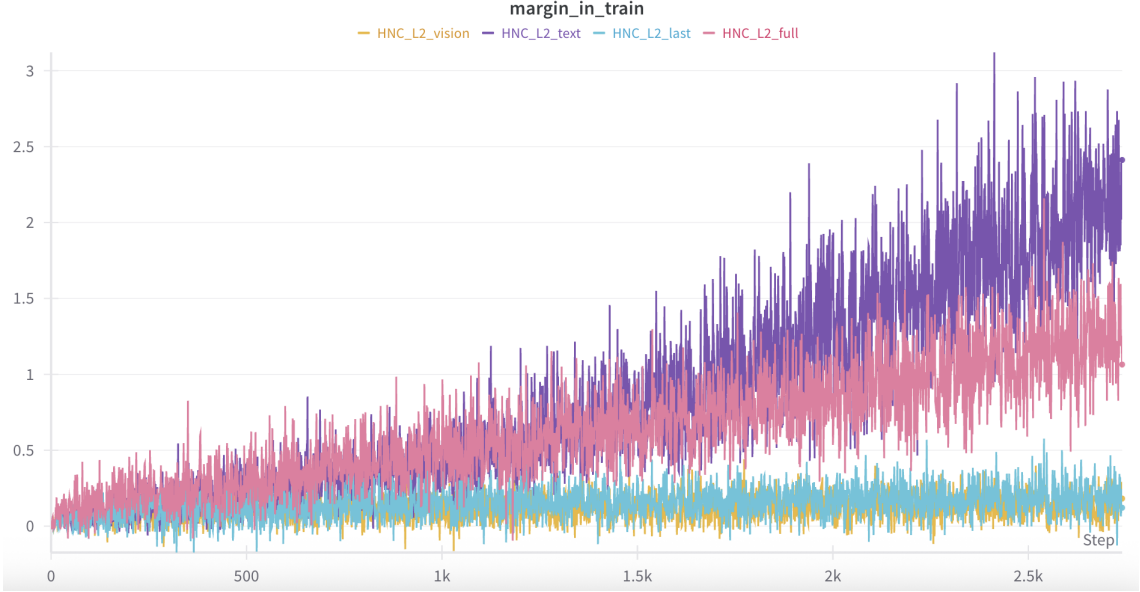


Figure 7: Margin growth during training across fine-tuning modes.

As shown in Figure 7, margin scores increase steadily for the `text_encoder` and `full_encoder` settings, indicating growing contrastive separation between positive and hard negative captions. Notably, the `text_encoder` variant achieves the largest margin throughout training, despite having lower positive scores than the `full_encoder` (as observed earlier). This further confirms the critical role of the text encoder in learning fine-grained semantic distinctions for this task. In contrast, the `vision_encoder` and `last_encoder` variants exhibit very limited growth in margin. This flat trend suggests that changing only the vision part of the model does not help much in telling apart captions with subtle meaning differences. Overall, these results show that adjusting the text side is especially important for learning relational information.

Table 2 reports the evaluation results on the HNC test set using the model checkpoints selected based on validation performance. Each column corresponds

Model	Avg_Pos	Avg_Neg	Margin	Rand_Neg	Acc@1
ViT-B/32 (Base)	0.2478	0.2457	0.0021	0.1769	0.5459
HNC_L2_text	0.2448	0.2396	0.0052	-0.0020	0.4587
HNC_L2_vision	0.3552	0.3465	0.0087	0.0035	0.5103
HNC_L2_last	0.4639	0.4515	0.0124	0.0277	0.5264
HNC_L2_full	0.5799	0.5636	0.0163	-0.0028	0.5287

Table 2: Test scores of different training parameter modes.

to a different evaluation metric: **Avg_Pos** is the average cosine similarity between each image and its positive caption; **Avg_Neg** is the similarity to its hard negative caption; **Margin** is the difference between these two scores, reflecting contrastive separation. **Rand_Neg** refers to similarity between each image and a randomly selected mismatched caption from the batch. **Acc@1** measures the proportion of samples where the model scores the positive caption at least 1.0 times higher than the hard negative caption, providing a threshold-based measure of semantic preference. The first row corresponds to the base CLIP model (ViT-B/32) without any fine-tuning. The remaining models are all fine-tuned using the HNC_L2 objective, which applies L2 regularization to constrain deviation from the original CLIP parameters. The suffixes in each model name denote the fine-tuned parameter strategy: **text** means only the text encoder is updated, as such. While tuning only the text encoder largely improves the model’s margin compared to the base model, it does not achieve the highest scores. Instead, the **full_encoder** configuration produces the strongest performance overall, with the best positive similarity, largest margin, and lowest random negative score. These results confirm the advantage of jointly fine-tuning both visual and textual components. As a result, the full encoder setup is used for all subsequent experiments to do further investigation and comparison.

To further examine the best-performing model, which is the fine-tuned full encoder model, Figure 8 plots the raw and scaled cosine similarity scores across training steps. Each panel tracks a similarity score of different caption type: positive, hard negative (HNC), and randomly sampled negative captions. This analysis is con-

ducted over an extended training duration of 50 epochs, with 137 steps per epoch, totaling 6850 training steps. This longer training schedule helps to better observe the trend of each score.

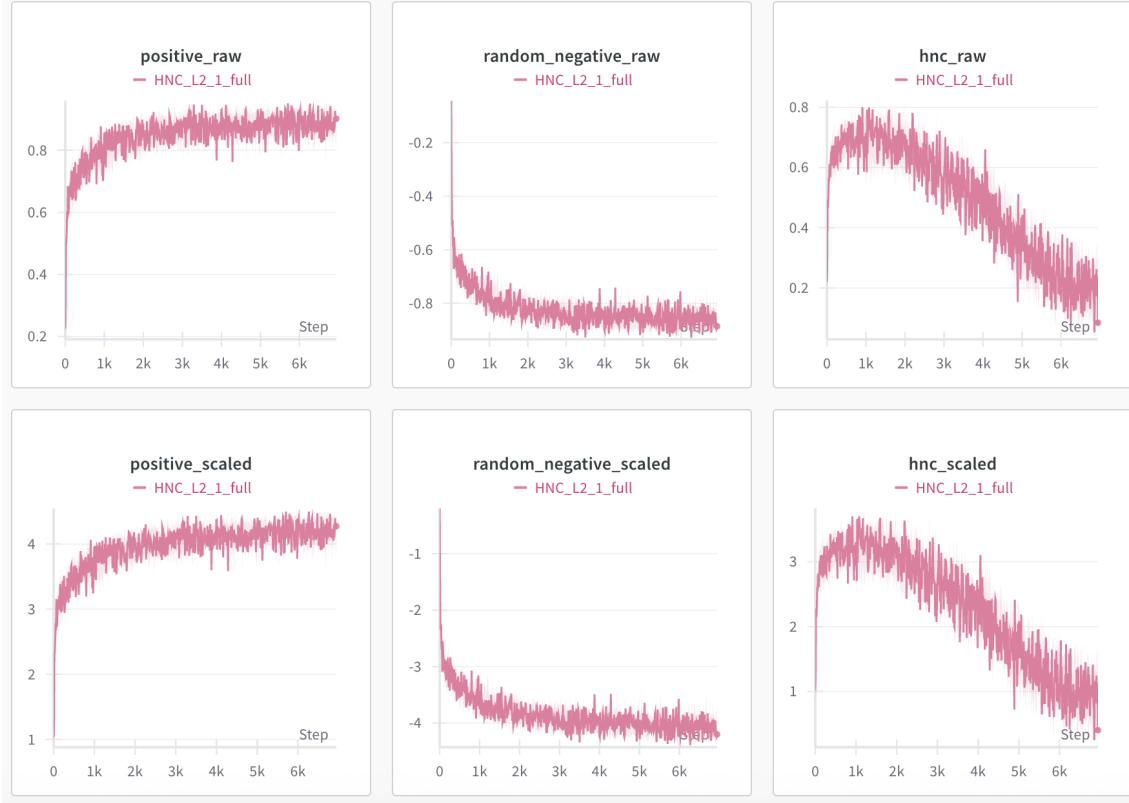


Figure 8: Training progression of cosine similarity scores (raw and scaled) for the full encoder model.

From the plots, the **positive** scores increased in both raw and scaled space, suggesting that the model learns to better align images with their true captions. In contrast, the **random negative** scores drop rapidly and stabilize at low values, reflecting strong contrastive learning pressure to separate mismatched pairs.

Interestingly, the **hard negative** scores exhibit a more nuanced trend. Initially, the scores rise as the model begins to capture general visual–textual associations. However, after approximately 2,000 steps, both raw and scaled HNC scores begin to

decline. This reversal indicates that the model gradually learns to suppress captions that are grammatically plausible but semantically misleading.

These trends provide a more fine-grained view of the training process and reinforce earlier observations that the `full_encoder` setting enables strong discriminative learning, not only between obvious mismatches but also for subtle relational differences.

5.2 Cosine Similarity Visualization

Motivated by the observation that both positive and hard negative similarity scores increase after fine-tuning, as shown in previous test results in Table 2, cosine similarity matrices are visualized to gain further insight. Specifically, the matrices display cosine similarity scores between image and their associated positive and hard negative captions. Each matrix is of size $N \times 2N$, where the first N columns represent positive captions and the second N represent hard negatives. These similarity scores are shown as heatmaps in Figure 9, allowing qualitative analysis of alignment patterns before and after fine-tuning.

In the base model visualization (top), the cosine similarity scores are relatively low and lack a clear diagonal structure. This indicates that the model does not effectively align images with their correct captions before fine-tuning. The expected strong diagonal, representing correct matches, is only faintly visible and blends into the background, suggesting insufficient discriminative alignment in the pretrained state.

After fine-tuning, the similarity matrix exhibits two clearly defined diagonal structures: the left diagonal corresponds to image–positive caption pairs, while the right diagonal represents image–hard negative caption pairs. Both diagonals are visually sharp and similarly intense, making it difficult to distinguish the absolute values by eye. The presence of a strong diagonal in the HNC block suggests that these captions, while semantically incorrect, still retain substantial image-relevant content. As a result, the HNC similarity scores are consistently higher than those of random mismatched captions.

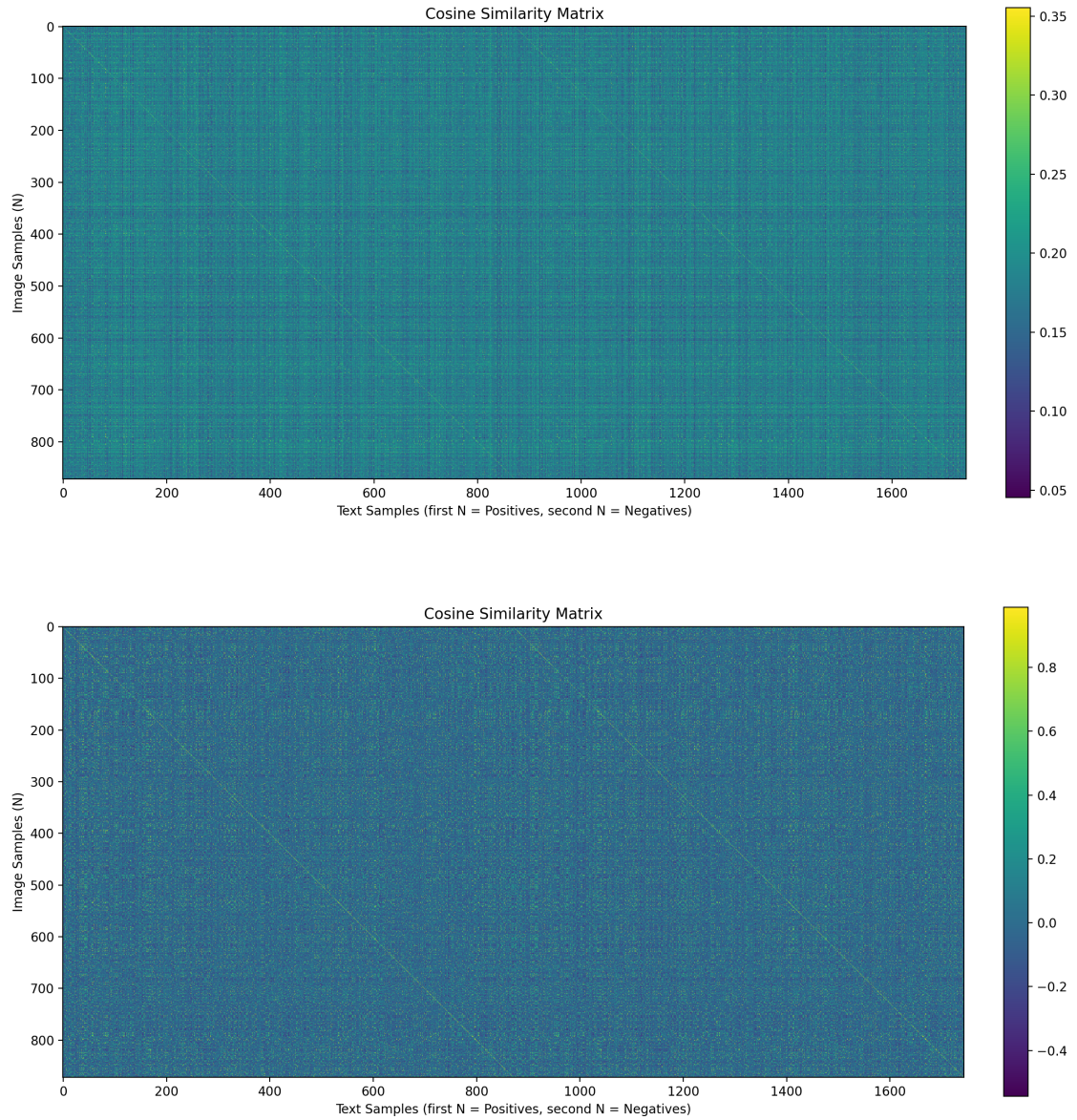


Figure 9: Cosine similarity matrices. Top: CLIP base model; Bottom: fine-tuned `full_encoder` model. Each matrix compares N images with $2N$ captions (first N positive, second N hard negatives).

In summary, the visualized similarity matrices offer a qualitative perspective on the model’s alignment behavior. The fine-tuned model exhibits clear diagonal patterns for both positive and hard negative captions. Although hard negatives still yield moderately high similarity scores, owing to their shared visual context, the overall contrastive structure becomes more pronounced.

5.3 Comparison of Regularization Strategies: L2 vs. KL

To investigate the effect of regularization on fine-tuning, two strategies are compared under the `full_encoder` configuration: L2 regularization and Kullback–Leibler (KL) divergence. Both are designed to mitigate catastrophic forgetting and preserve zero-shot objective detection ability of the original CLIP model.

Metric	Base	L2_text	L2_vision	L2_last	L2_full	KL_full
Avg_Pos	0.2478	0.2448	0.3552	0.4639	0.5799	0.6023
Avg_Neg	0.2457	0.2396	0.3465	0.4515	0.5636	0.5779
Margin	0.0021	0.0052	0.0087	0.0124	0.0163	0.0244
Rand_Neg	0.1769	-0.0020	0.0035	0.0277	-0.0028	0.0093
Acc@1	0.5459	0.4587	0.5103	0.5264	0.5287	0.5631
Acc@1.1	0.1227	0.2695	0.2489	0.2592	0.2993	0.2672
Acc@1.2	0.0470	0.1927	0.1755	0.1778	0.2167	0.1927
Acc@1.5	0.0000	0.1261	0.0860	0.0963	0.1227	0.1055
Acc@2	0.0000	0.0849	0.0482	0.0550	0.0711	0.0608
Acc@3	0.0000	0.0493	0.0241	0.0310	0.0333	0.0252

Table 3: Evaluation results across multiple metrics on the human-annotated HNC test dataset.

Table 3 extends the previous comparison by including results from the `KL_full` model, which applies KL divergence regularization during full encoder fine-tuning. As before, the metrics cover average cosine similarity scores with positive, hard negative, and randomly mismatched captions, the margin, and threshold-based accuracy

(**Acc@1** to **Acc@3**). This allows for a direct comparison between L2 and KL regularization strategies under otherwise consistent training settings on the manually annotated HNC test dataset.

Compared to the `L2_full` model, the `KL_full` variant achieves slightly higher average similarity scores with both positive (0.6023 vs. 0.5799) and hard negative captions (0.5779 vs. 0.5636), resulting in the highest overall margin (0.0244). Additionally, the KL-regularized model achieves the highest **Acc@1** score (0.5631), outperforming both the base CLIP model and the L2-regularized variant. This improvement indicates a stronger ability to identify the correct caption among each pair. However, under other thresholds such as **Acc@1.1** and **Acc@1.2**, the L2 variant slightly outperforms KL, suggesting that L2 may facilitate better contrastive discrimination under stricter semantic thresholds.

Overall, KL divergence regularization demonstrates strong potential in preserving pretrained alignment while enhancing fine-grained relational sensitivity. Nevertheless, the choice between L2 and KL may depend on the evaluation criteria: KL performs best under relaxed thresholds (e.g., **Acc@1**), indicating stronger general alignment, whereas L2 outperforms KL under stricter thresholds, suggesting greater robustness in detecting subtle semantic differences.

5.4 Performance by Semantic Type

To better understand the model’s ability to handle different semantic phenomena, performance is broken down by caption type. The HNC test set includes 10 categories such as relational reasoning, logical operations (e.g., **xor**, **and**), attribute verification, and object counting or comparison. This breakdown helps reveal which reasoning skills are strengthened through fine-tuning and which remain challenging.

Figure 10 shows the composition of the test dataset after filtering. The most common type is **relation**, followed by various logic-based, attribute-based, and counting-related categories. The figure illustrates the relative distribution of each type among the 872 total samples.

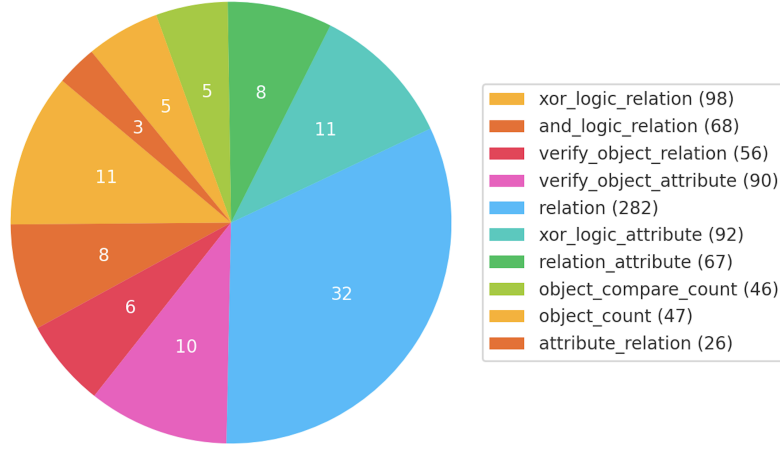


Figure 10: Distribution of semantic types in the HNC test dataset after filtering.

Type	Samples	Base	KL_full	Improvement
ALL	872	0.0021	0.0244	0.0223
relation	282	0.0074	0.0574	0.0500
xor_logic_relation	98	-0.0045	-0.0094	-0.0049
xor_logic_attribute	92	-0.0042	-0.0056	-0.0014
verify_object_attribute	90	0.0024	0.0302	0.0278
and_logic_relation	68	0.0029	-0.0133	-0.0162
relation_attribute	67	0.0023	0.0220	0.0197
verify_object_relation	56	0.0003	0.0126	0.0123
object_count	47	0.0003	0.0433	0.0430
object_compare_count	46	0.0000	-0.0041	-0.0041
attribute_relation	26	0.0011	0.0262	0.0251

Table 4: Margin Comparison by Type.

Table 4 reports the average margin for each semantic type in the HNC test dataset, comparing the base CLIP model (ViT-B/32) and the fine-tuned model. The final column shows the margin improvement after fine-tuning.

From the table we can see, the fine-tuned model achieves margin gains on most types, especially in tasks involving grounded visual understanding. The **relation** (+0.0500), **object_count** (+0.0430), and **verify_object_attribute** (+0.0278) types benefit the most from training, suggesting that the model becomes more sensitive to spatial and object-level semantics through contrastive fine-tuning.

Moderate gains are also observed in **attribute_relation**, **relation_attribute**, and **verify_object_relation**, indicating improved relational reasoning when the linguistic structure remains relatively simple.

Figure 11 provides a visual comparison of margin scores across types. It clearly shows large improvements for these grounded types, while performance remains flat or even declines for logic-intensive categories.

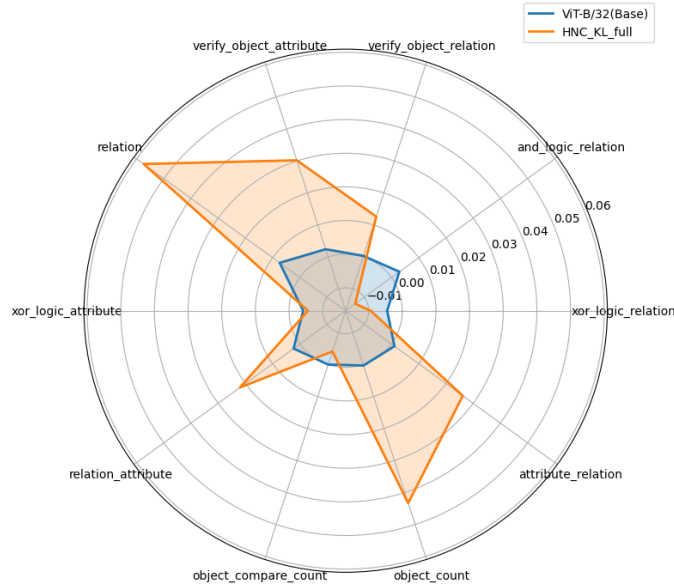


Figure 11: Radar plot comparing margin scores by type for base model and KL-regularized full encoder fine tuned model.

Logic-heavy types such as `and_logic_relation`, `xor_logic_relation`, `object_compare_count`, and `xor_logic_attribute` show minimal or negative improvement. These types often require abstract reasoning about logical operators or quantity comparisons, which may be harder to learn from paired image-text supervision alone.

To illustrate this, Figure 12 shows a sample image and caption pairs corresponding to the most difficult logic-based types. These examples emphasize the subtle changes in semantics that the model finds difficult to resolve:



Figure 12: Example image with logic-related captions from difficult semantic types.

Type - `and_logic_relation`

Pos: There are both a lamp behind the women and a laptop in front of the window.

Neg: There are both a lamp in front of the women and a laptop in front of the window.

Type - `xor_logic_relation`

Pos: The laptop is either behind the women or the sofa.

Neg: The laptop is either behind the women or the foot.

Type - `object_compare_count`

Pos: There are as many remote controls as people.

Neg: There are fewer remote controls than people.

Type - xor_logic_attribute

Pos: There is either a red carpet or green curtains.

Neg: There is either a red carpet or purple curtains.

In summary, the analysis shows that fine-tuning significantly enhances the model’s ability to handle relational and attribute-based reasoning. However, abstract logic remains a persistent weakness, highlighting an area for potential future work such as integrating more structured supervision.

5.5 Generalization to COCO-Derived Test Set

To evaluate the generalization ability of the model beyond the HNC dataset, a new test set was constructed by randomly sampling 1,000 Image–Positive–HNC triples from the COCO-derived dataset.

Metric	Base	L2_full	KL_full
Avg_Pos	0.3043	0.7198	0.7407
Avg_Neg	0.2973	0.7020	0.7255
Margin	0.0071	0.0178	0.0151
Avg_Rand_Neg	0.1510	0.0265	0.0231
Acc@1	0.7160	0.6180	0.6290
Acc@1.1	0.0750	0.1740	0.1170
Acc@1.2	0.0090	0.0930	0.0630
Acc@1.5	0.0000	0.0320	0.0160
Acc@2	0.0000	0.0150	0.0080
Acc@3	0.0000	0.0060	0.0000

Table 5: Performance on COCO-derived test set.

Table 5 summarizes the performance. Both fine-tuned models (`L2_full` and `KL_full`) show a substantial increase in average positive similarity, improving from 0.30 in the base model to over 0.71 and 0.74 respectively. These gains suggest that fine-tuning significantly enhances the model’s ability to match images with relevant captions, even outside the training distribution.

The margin between positive and hard negative scores also increases, with the `L2_full` model achieving the highest margin (0.0178), followed by `KL_full` (0.0151), and the base model showing only 0.0071. This reflects improved contrastive separation after fine-tuning.

Interestingly, `Acc@1` is highest in the base model (0.7160), slightly outperforming the fine-tuned models. This shift suggests that while relational fine-tuning improves semantic contrast and alignment precision, it may also slightly reduce the model’s general image–text matching performance. Nonetheless, at more strict thresholds (e.g., `Acc@1.1–Acc@2`), both fine-tuned models outperform the base, especially `L2_full`, which achieves the highest scores across most metrics.

In summary, fine-tuning improves the model’s semantic alignment and margin scores even on a COCO-derived test set that differs from the original HNC construction. While accuracy at the `Acc@1` threshold drops slightly, indicating reduced tolerance for broadly relevant captions, the fine-tuned models exhibit more robust discrimination under stricter evaluation criteria. These results suggest that fine-tuning strengthens the model’s ability to capture relational details, even if this comes at the cost of reduced flexibility in handling loosely aligned captions. Still, the observed gains in alignment and margin demonstrate strong generalization beyond the training domain.

5.6 Limitation in Large Hard Negative Weighting

An important design choice in training is how much emphasis to place on hard negative captions during optimization. This is controlled by the HNC weight hyperparameter, which scales the loss contribution from hard negatives. All previous re-

sults were obtained using a weight of 1.0, but additional experiments were conducted to evaluate the effect of larger or dynamically adjusted weights.

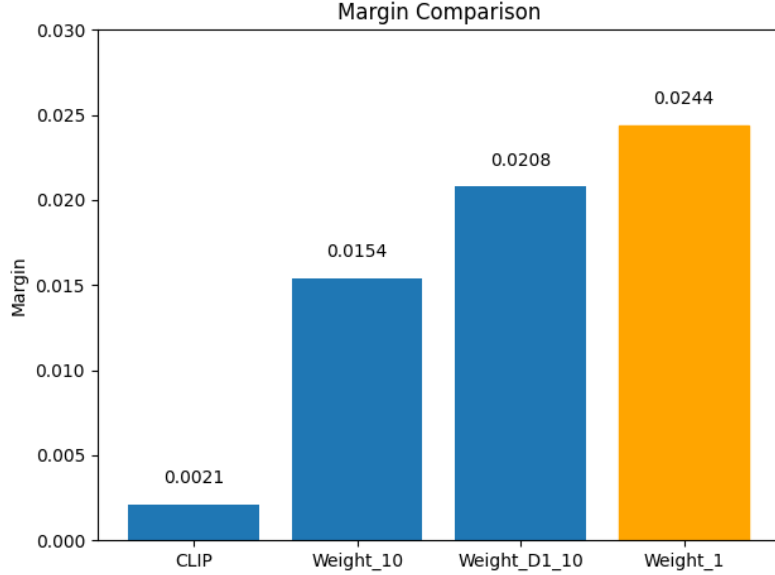


Figure 13: Comparison of margin scores for different weighting strategies.

Figure 13 compares the resulting margin scores for several configurations. The CLIP baseline (no fine-tuning) yields a very small margin of 0.0021, indicating poor separation between positive and hard negative captions. When using a fixed HNC weight of 10 (denoted **Weight_10**), the margin improves to 0.0154, but the gain is limited. A dynamic strategy (**Weight_D1_10**) that starts from weight 1 and gradually increases to 10 over approximately the first 1,000 training steps, then remaining fixed at 10 for the remainder of training, visualized in Figure 14. It achieves a slightly higher margin of 0.0208, and suggests some benefit from incrementally introducing HNC emphasis rather than applying strong pressure from the beginning. However, the best result (0.0244) comes from the simple fixed-weight setting of 1.0 (**Weight_1**), which balances learning difficulty and stability during training.

These results indicate that larger HNC weights do not always lead to better performance. One likely reason is that hard negatives are especially difficult, if their contribution dominates early training, the model may struggle to form effective alignment. The dynamic strategy offers some mitigation by postponing their impact,

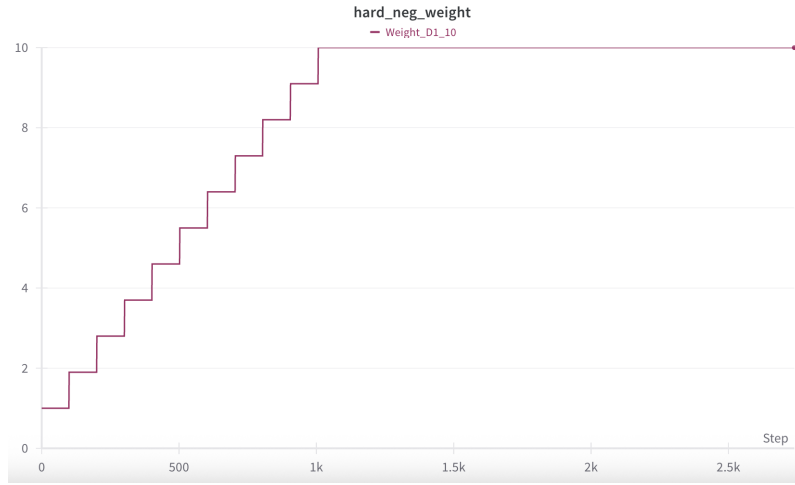


Figure 14: Schedule of dynamic HNC weight increase in the `Weight_D1_10` configuration.

but still falls short of the default setting. This highlights a limitation of overly aggressive hard negative training and emphasizes the importance of proper weighting and training schedule design.

6 Explanation

To provide a more intuitive explanation of how the model interprets image–caption relationships, this section applies the explanation framework, which introduces a second-order attribution method for CLIP-based models. This method decomposes the final similarity score into fine-grained contributions from image-text feature pairs, capturing cross-modal interactions between specific words and regions.

6.1 Explanation Pipeline

To better visualize how CLIP models understand the connection between images and captions, I adopt a second-order attribution method proposed by Möller and Tilli et al. (Möller et al., 2025). This method provides insights into how individual words in a caption interact with specific regions in an image, which first-order methods, such as Grad-CAM (Selvaraju et al., 2019) or Integrated Gradients (Sundararajan et al., 2017), cannot capture well, because CLIP uses separate encoders for text and image, and the similarity score depends on interactions between both.

The method proceeds in three main steps. First, given an image and a caption, the CLIP model encodes each input independently into an embedding. Their similarity is then computed as the dot product or cosine similarity between these embeddings.

Second, to attribute this similarity to interactions between specific words and image regions, the method computes a second-order attribution tensor. This tensor approximates the joint contribution of each word–patch pair by integrating the mixed partial derivatives of the similarity function with respect to both the image and text inputs, along linear paths from uninformative reference inputs to the actual ones. After summing over the embedding dimensions, this yields a three-dimensional tensor of shape $H \times W \times S$, where each value reflects the estimated contribution of a specific image patch and caption token pair.

Third, this tensor is projected along one axis for visualization. Summing across the token dimension gives a spatial heatmap over the image, showing where the

model grounds specific words. Conversely, summing over the image patches yields word-level relevance scores, indicating which parts of the caption contribute most to a selected image region.

Unlike methods that treat the image and caption separately, this second-order technique directly attributes the output to specific interactions. As a result, it gives us a clearer picture of how CLIP aligns language and vision at a detailed level. In the next subsection, I use this method to compare base and fine-tuned CLIP models on a simple relational example.

6.2 Attribution Visualization Example

To more intuitively understand how fine-tuning affects the model’s interpretation of image–caption relationships, I apply the second-order explanation pipeline introduced earlier to both the base CLIP model and the fine-tuned model. This visualization includes two components: a text-to-image heatmap showing which parts of the image are important for understanding the caption, and an image-to-text attribution map highlighting which words are most influenced by the image.

To conduct this analysis, I use the following relational caption, which involves a spatial relation, making it an ideal candidate to examine whether the model captures not just object presence but also their relationships:

The trees are behind the fence

Figure 15 highlights which parts of the image the model considers most relevant when computing the similarity score. The first panel shows the original image, used as input to both models. The middle panel displays the attribution heatmap produced by the base CLIP model, while the right panel shows the results after fine-tuning with relational supervision.

In both heatmaps, the colors represent the estimated contribution of each image region to the final similarity score between the image and the caption: red indicates high contribution, while green indicates relative weak contribution and blue indicates



Figure 15: Comparison of text-to-image heatmap between base CLIP and fine-tuned model for the caption ‘The trees are behind the fence’.

none or negligible influence. These maps reveal where the model focuses to justify its alignment between the image and the caption.

In the base CLIP model, contribution is diffuse and misdirected, mainly focused on irrelevant areas like the “Grand Opening” sign in the foreground. This suggests the model struggles to connect the correct spatial and object cues described in the text.

By contrast, the fine-tuned model shifts its contribution toward more semantically relevant regions. The trees in the background and the fence between them. These areas receive strong red and yellow activations, indicating a better grounding of visual evidence that supports the relational phrase “behind the fence.” This demonstrates that fine-tuning enhances the model’s ability to connect language and vision in a meaningful way.

To gain deeper insight into how specific words are grounded in the image, I compare token-level visual attribution maps across the base and fine-tuned CLIP models. Figures 16, 17, and 18 illustrate the relevance heatmaps for the words “trees” “fence” and “behind” respectively. Each pair of subfigures contrasts the base model (left) and the fine-tuned model (right).

In Figure 16, the base model’s attribution for the token “trees” is weak and dispersed, with some irrelevant attention falling on the ramp structure. In contrast, the fine-tuned model can mainly find the trees’ location in the image despite some noise. It shows strong and focused activation on the actual trees in the background.

This indicates improved object localization after fine-tuning.

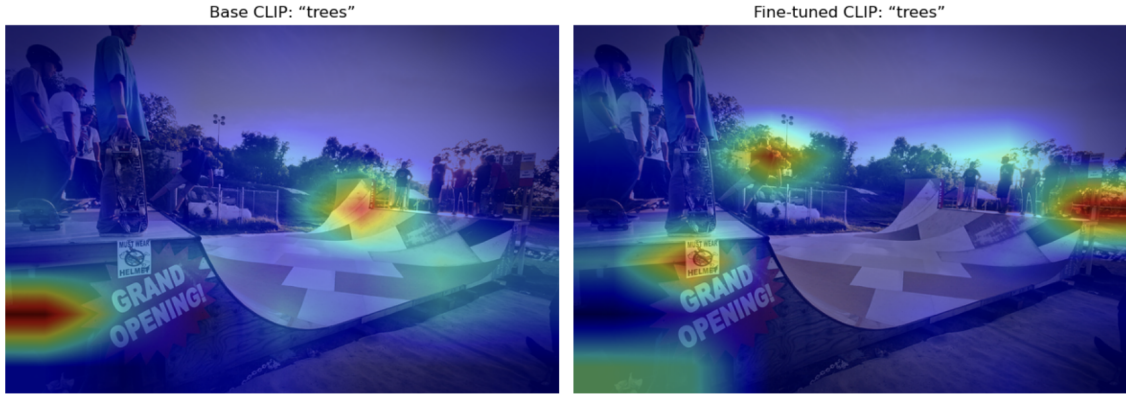


Figure 16: Relevance for token “trees” across base and fine-tuned models.

Figure 17 shows a similar improvement for the word “fence”. While the base model mostly attends to the left side of the image with limited relevance on the actual fence, the fine-tuned model highlights the entire fence line more clearly and comprehensively. This suggests that the model has learned to associate the term “fence” with the correct visual evidence.



Figure 17: Relevance for token “fence” across base and fine-tuned models.

Finally, Figure 18 illustrates the relevance for the relational word “behind”. The base model again focuses incorrectly on the center of the image, far from the region relevant to spatial reasoning. In contrast, the fine-tuned model shifts attention

toward both the tree and fence areas, effectively capturing the spatial relationship encoded by “behind”.



Figure 18: Relevance for token “behind” across base and fine-tuned models.

Together, these visualizations demonstrate that fine-tuning not only strengthens the model’s ability to recognize objects but also improves its understanding of how those objects relate to one another in the scene. This supports the conclusion that relational understanding in CLIP can be meaningfully improved through targeted supervision of semantic relationships.

In addition to looking at where the model focuses in the image, I also analyze which words in the caption are most influenced by the image. Figure 19 shows the token-level image-to-text attribution visualization, comparing the base CLIP model and the fine-tuned model. The color intensity reflects how much each word contributes to the final image–caption similarity score, with red indicating positive contributions and blue indicating negative ones.

In the base CLIP model, the tokens “the” and “trees” are shaded blue, meaning they contribute negatively to the similarity score. Only “fence” receives a strong positive attribution, while “behind” receives some attention but remains relatively weak.

After fine-tuning, all tokens shift to having positive attribution, indicating better alignment between the text and visual features. The word “trees” now receives the strongest emphasis, while “fence” and “behind” also contribute positively, though to

a lesser extent. This shows that the fine-tuned model has learned to ground more of the caption content, resulting in a more holistic understanding of the image-caption pair.

Base CLIP

the trees are behind the fence

Fine-tuned CLIP

the trees are behind the fence

Figure 19: Image-to-text word attribution comparison.

Taken together, these visualizations give an interpretation of improvement in fine-tuned model. The model learns to better associate specific image regions with corresponding textual elements and becomes more sensitive to subtle semantic distinctions. Notably, even relational words like "behind" are better grounded visually in the fine-tuned model.

Overall, this explanation framework provides a useful lens to examine how relational fine-tuning affects not just final scores, but also how meaning is represented and distributed internally. It highlights the importance of visual grounding for compositional semantics and supports the conclusion that fine-tuned CLIP models achieve more coherent and interpretable alignment.

7 Conclusions and Future Work

In this thesis, I proposed an effective method to enhance the relational understanding capabilities of the CLIP model through the integration of Hard Negative Captions. While CLIP achieves strong performance on a broad range of vision–language tasks, its reliance on global image–text alignment makes it less effective at capturing fine-grained relational semantics, such as spatial arrangements, role assignments, and logical structures.

To address this limitation, I designed a modified contrastive learning framework that incorporates HNC supervision, grammatically correct but semantically contradictory captions that are constructed to target relational distinctions. This approach extends the traditional CLIP training objective by weighting these hard negatives, either statically or dynamically, within the loss function. To preserve CLIP’s generalization ability, I also applied regularization through either L2 penalties or KL divergence to the pretrained model.

Experiments across multiple configurations demonstrated that relational understanding can be substantially improved through targeted fine-tuning. The results showed a consistent increase in the margin between positive and hard negative captions, indicating that the model became more sensitive to subtle semantic differences. Among the encoder configurations, tuning the text encoder played a particularly important role in improving semantic discrimination, while full encoder tuning achieved the strongest performance overall across similarity metrics, margin, and accuracy.

In terms of regularization, KL divergence yielded the highest margin scores, suggesting better contrastive precision, whereas L2 regularization showed superior robustness under stricter accuracy thresholds. Importantly, these improvements were not limited to the original dataset. On a COCO-derived test set, which constructed independently of the HNC corpus, the fine-tuned models retained their relational advantage, indicating strong generalization to new domains.

Beyond numerical gains, interpretability also improved. Contribution heatmaps revealed that the fine-tuned model better localized visual evidence corresponding to

both object terms and relational words, such as “behind,” suggesting that relational concepts became more grounded in visual features.

Despite the encouraging outcomes, this study also revealed several important limitations. Although training with Hard Negative Captions improved the model’s sensitivity to relational differences, its performance remains limited when handling logically complex or deeply nested captions. In sentences involving coreference structures, or hierarchical relationships, the model often fails to fully capture the intended semantics, suggesting that the relational understanding learned remains shallow in such cases. Addressing these limitations will require more advanced training strategies.

Second, I explored the use of dynamic weighting strategies to emphasize hard negative captions during training. While increasing their contribution was hypothesized to improve learning, the results showed the opposite: larger weights led to a degradation in performance, likely due to optimization instability. This highlights the need for a more principled exploration of weighting schedules. In future work, a more systematic investigation into the design of hard negative weights could offer deeper insights into their impact on training dynamics and relational discrimination.

Another extension involves applying the fine-tuned model to downstream tasks such as Visual Question Answering (VQA), image captioning, or referring expression comprehension. These applications provide a real-world evaluation of whether the relational understanding acquired through HNC training generalizes to more complex multimodal reasoning scenarios. Furthermore, performance on such tasks can offer insights into how relational semantics interact with pragmatic and commonsense reasoning in natural settings.

In addition, future work could explore the effects of fine-tuning hyperparameters more systematically—particularly the strength of regularization components such as L2 loss or KL divergence. This thesis applied fixed values, but adjusting these values conditionally may lead to better stability and performance.

In summary, this thesis presents a conceptually simple yet effective extension to the CLIP training paradigm. By injecting relationally informed supervision through

hard negative captions, I demonstrate that fine-tuning can substantially improve semantic sensitivity, model interpretability, and generalization across domains, bringing vision–language systems one step closer to structured, human-aligned understanding.

The full implementation are publicly released to promote reproducibility and future research.³

³https://github.com/WenW186079/HNC_CLIP_masterthesis

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